

约束满足问题

祝恩
计算机学院
国防科技大学



学习内容

1. 定义约束满足问题：

再看n-queens, CSP形式化, CSP实例, 变量与约束类型, 约束图, 约束超图

2. 约束传播(CSP中的推理)：

结点相容、弧相容、路径相容、K相容

1 定义约束满足问题

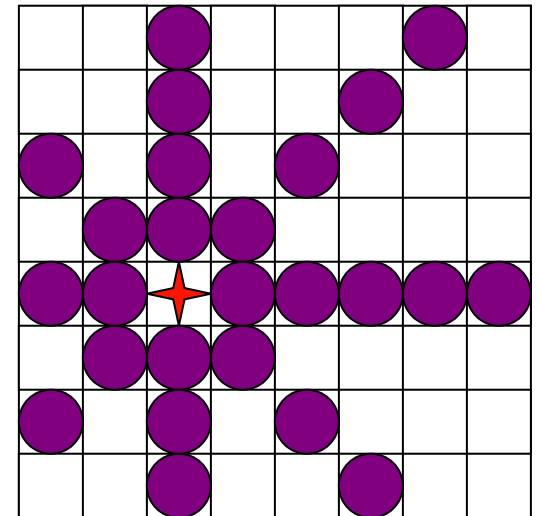


再看n-queens、CSP形式化、约束类型、约束图、约束超图

约束传播

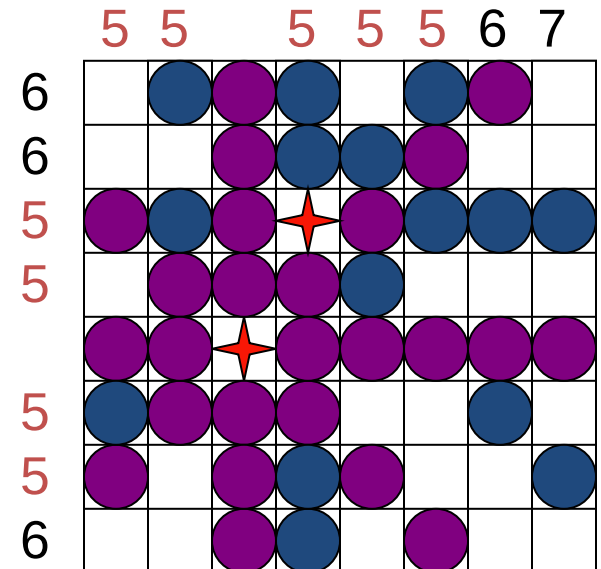
Constraint Propagation

- **Place** a queen in a square
- **Remove** the attacked square from future consideration



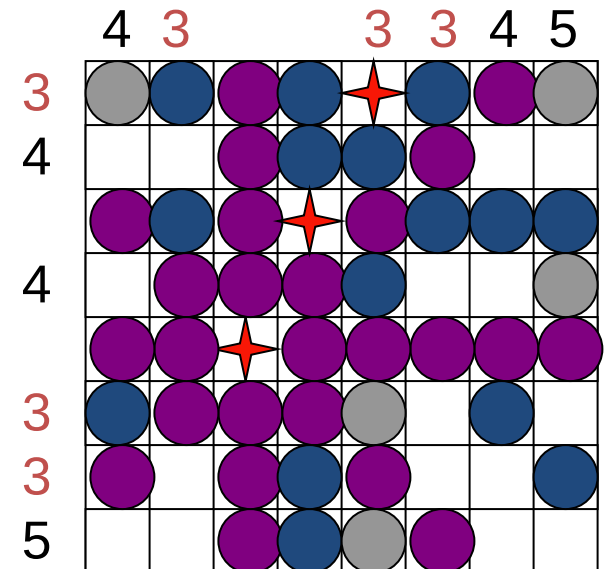
约束传播

- Count the number of non-attacked squares in every row and column
- **Place** a queen in a row or column with **minimum** number
- **Remove** the attacked squares from future consideration



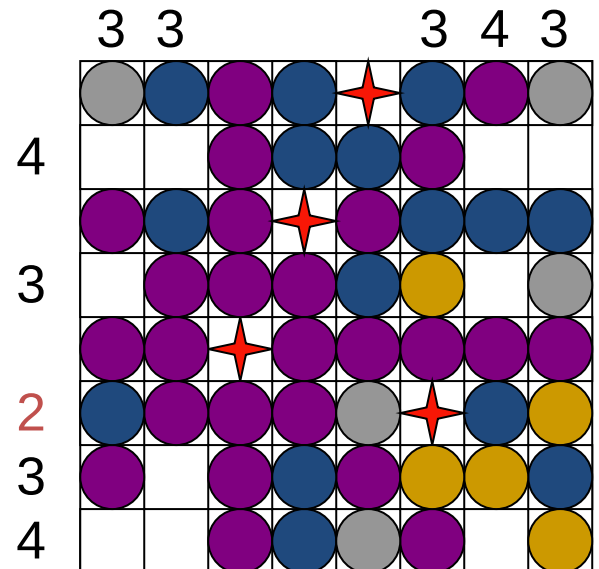
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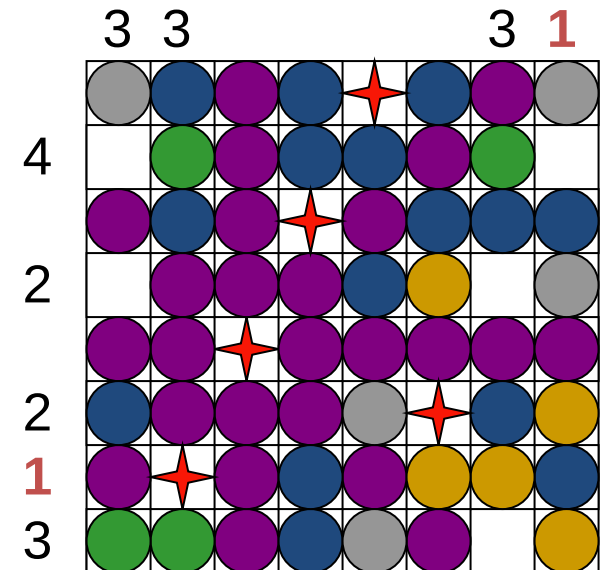
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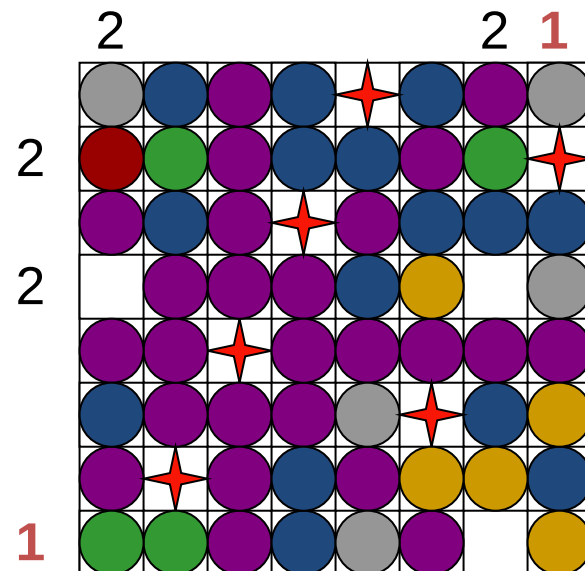
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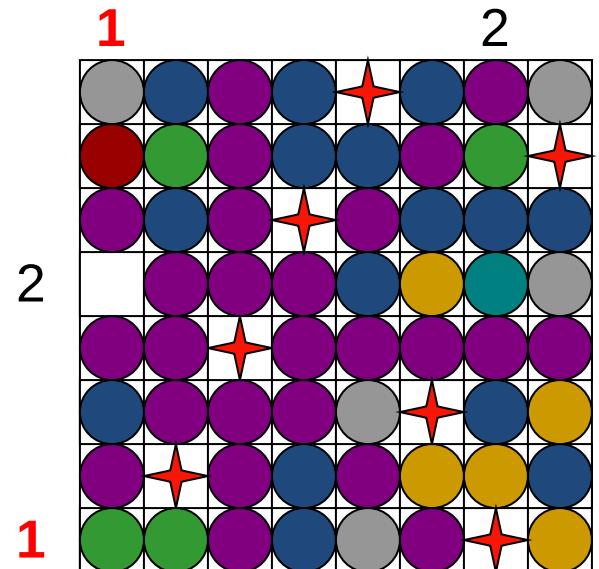
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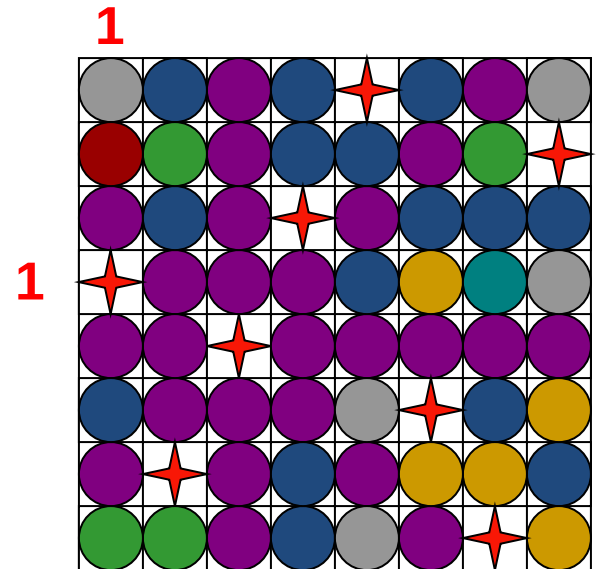
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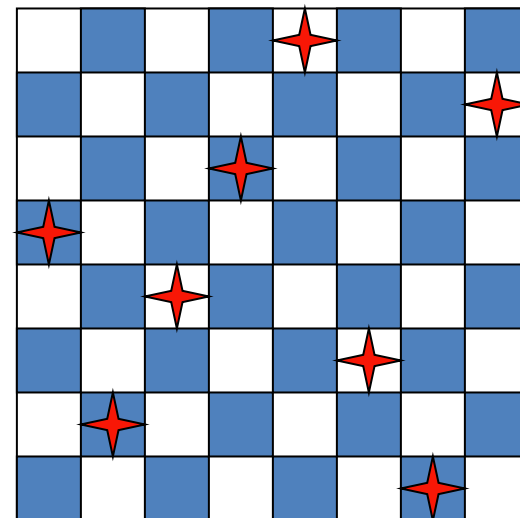
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约束传播

- Count the number of non-attacked squares in every row and column
- Place a queen in a row or column with minimum number
- Remove the attacked squares from future consideration



约束传播 vs. 局部搜索

- local search: complete assignment
- constraint propagation: partial assignment

定义约束满足问题

- **约束满足问题**：使用一组变量(要素化表示)来描述状态，当每个变量都有自己的赋值，且赋值满足所有关于变量的约束时，问题就得到了解决。简称CSP。



形式化CSP

CSP的形式化

□ a CSP consists of

■ a set of variables, $X = \{X_1, \dots, X_n\}$.

■ a set of domains, $D = \{D_1, \dots, D_n\}$.

■ a set of constraints, C .

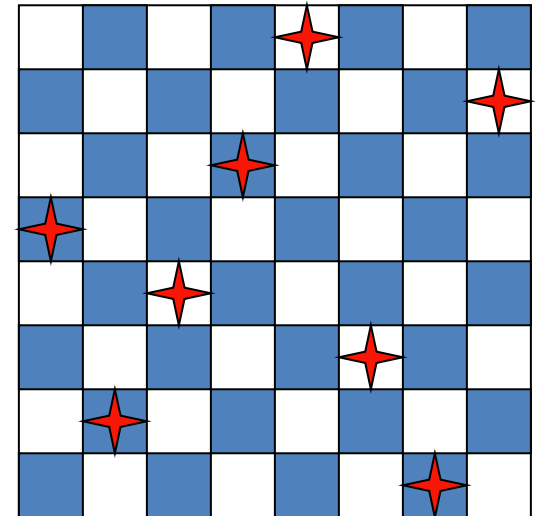
Each constraint C_i consists of $\langle \text{scope}, \text{rel} \rangle$.

scope is a tuple of variables, and *rel* defines the relation between those variables.

□ Solutions are complete and consistent assignments

8-queens的CSP形式化

- CSP formulation of 8-queens
- Variables: $X=\{X_1,X_2,X_3,...,X_8\}$
- Domain: $D=\{D_1,D_2,D_3,...,D_8\}$
 $D_i=\{1,2,3,4,5,6,7,8\}$
- Constraints: $C=\{<\{X_i,X_j\}, X_i \neq X_j \ \& \ |X_i-X_j| \neq |i-j| \mid i \neq j, 1 \leq i \leq 8, 1 \leq j \leq 8\}$
- a solution $\{5,2,4,6,8,3,1,7\}$



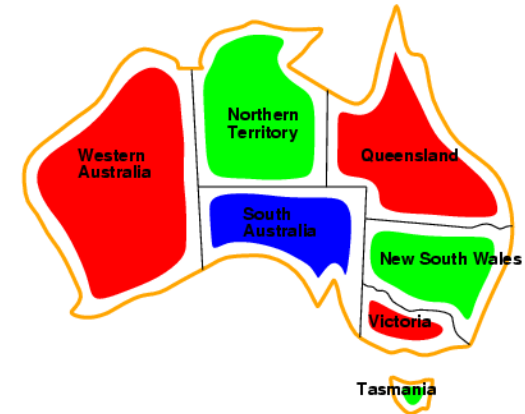
地图着色的CSP形式化

- Variables: $X = \{WA, NT, Q, NSW, V, SA, T\}$
- Domain: $D = \{D_1, \dots, D_7\}, D_i = \{red, green, blue\}$
- Constraints: adjacent regions must have different colors
 $C = \{WA \neq NT, WA \neq SA, NT \neq SA, NT \neq Q, SA \neq Q, SA \neq NSW, SA \neq V, Q \neq NSW, NSW \neq V\}$

$SA \neq WA$ 是 $\langle (SA, WA), SA \neq WA \rangle$ 的简洁表示

- Solutions are assignments satisfying all constraints

$\{WA = red, NT = green, Q = red, NSW = green, V = red, SA = blue, T = green\}$



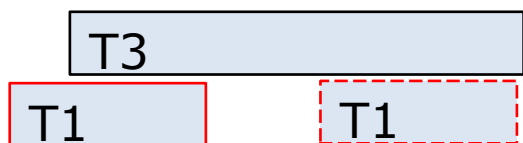
任务调度的CSP形式化

- Four tasks T1, T2, T3, and T4 are related by time constraints:
 - T1 must be done during T3
 - T2 must be achieved before T1 starts
 - T2 must overlap with T3
 - T4 must start after T1 is complete
- Their durations are D1,D2,D3,D4.
- How to formulate this problem as a CSP?
 - Variables
 - Domains
 - Constraints



任务调度的CSP形式化

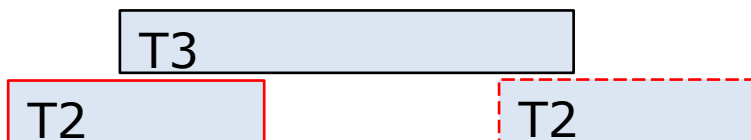
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$$T3 < T1 + D1, T1 + D1 < T3 + D3$$



$$T2 + D2 < T1$$



$$T2 + D2 \geq T3, T2 \leq T3 + D3$$



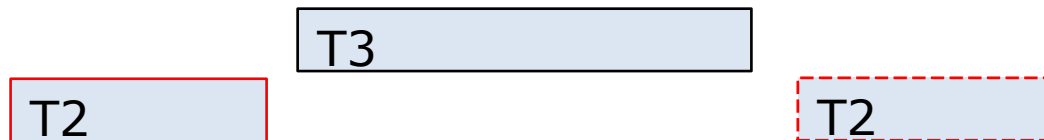
$$T4 > T1 + D1$$

任务调度的CSP形式化

- Four tasks T1, T2, T3, and T4 are related by time constraints:
 - T1 must be done during T3
 - T2 must be achieved before T1 starts
 - T2 must overlap with T3
 - T4 must start after T1 is complete

考虑T2和T3使用同一资源s, s只有一个, 每次只能供一个任务使用

T2 must not overlap with T3



$T2 + D2 < T3 \vee T3 + D3 < T2$

任务调度的CSP形式化

- Four tasks T1, T2, T3, and T4 are related by time constraints:
 - T1 must be done **during** T3
 - T2 must be achieved **before** T1 starts
 - T2 must **overlap** with T3
 - T4 must start **after** T1 is complete
- How to formulate this problem as a CSP?
- What are the possible time **relations** between two tasks?
- What if the tasks use **resources** in limited supply?
- Are the constraints **compatible**?



变量的类型

□ Discrete Variables

- Finite domains: Size d , $O(d^n)$ complete assignments
 - Boolean CSPs, including Boolean satisfiability (NP-complete)
- Infinite domains (integers, strings, etc.)
 - Need a constraint language
 - $\text{StartJob}_1 + 5 < \text{StartJob}_3$
 - Linear constraints solvable, nonlinear undecidable

□ Continuous variables

- start/end times for Hubble Telescope observations
- Linear programming: linear constraints solvable in polynomial time

约束的类型

- **Unary** constraints involve a single variable:

$SA \neq green$

- **Binary** constraints involve pairs of variables:

$SA \neq WA$

- **Higher-order** constraints involve 3 or more variables. $between(x,y,z)$

- **Global constraints** involve an arbitrary number of variables. $Alldiff(x,y,z,u)$

- **Preferences** (soft constraints)

- red is better than green

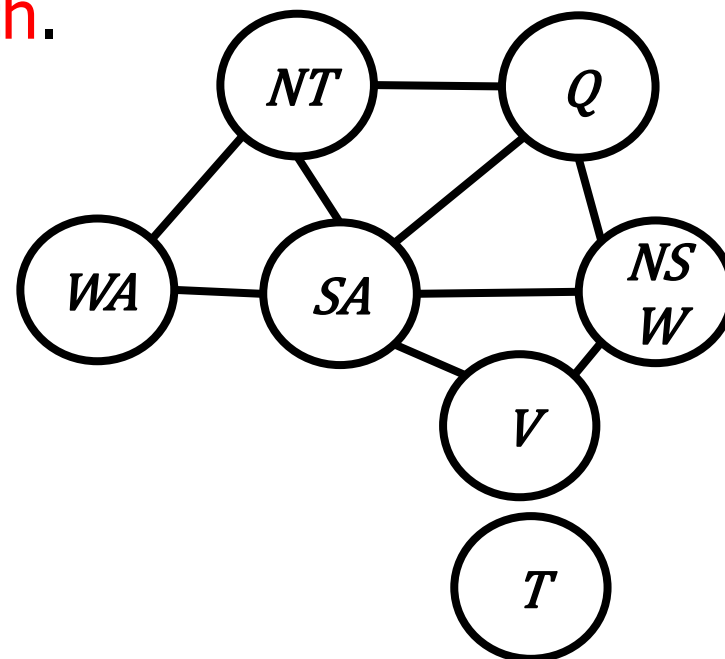
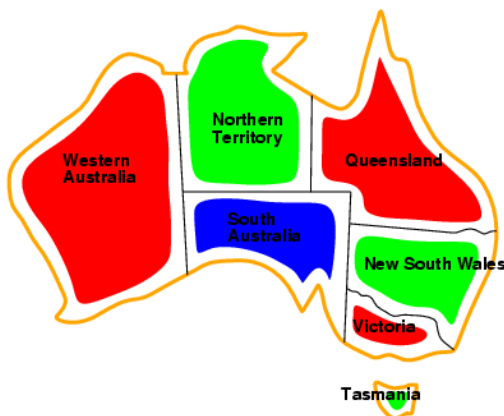
- Gives constrained optimization problems

- Representable by a cost for each variable assignment

绝对约束：任何违反约束的都被排除在解之外
偏好约束：指出哪些解更好——约束优化问题

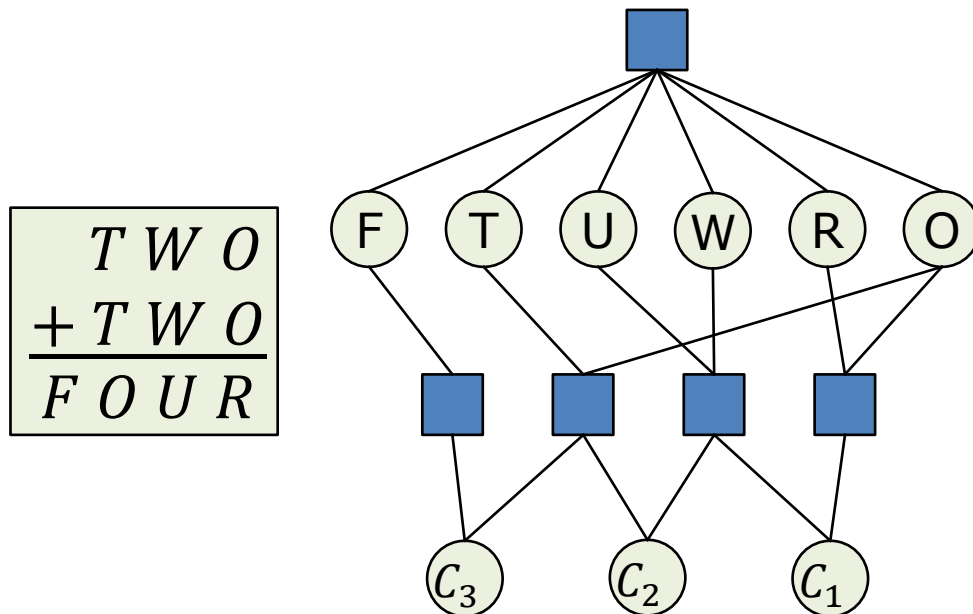
约束图

- ❑ **Binary CSP**: each constraint relates (at most) two variables
- ❑ Constraint graph: **nodes are variables, arcs show constraints**
- ❑ General-purpose CSP algorithms **use the graph structure to speed up search.**



约束超图

- 变量个数任意的约束称为全局约束，但不一定涉及所有变量。例如Alldiff
- 约束超图：表示高阶约束



$$\begin{aligned}
 &Alldiff(F, T, U, W, R, O) \\
 &O + O = R + 10 \cdot C_1 \\
 &C_1 + W + W = U + 10 \cdot C_2 \\
 &C_2 + T + T = O + 10 \cdot C_3 \\
 &C_3 = F
 \end{aligned}$$

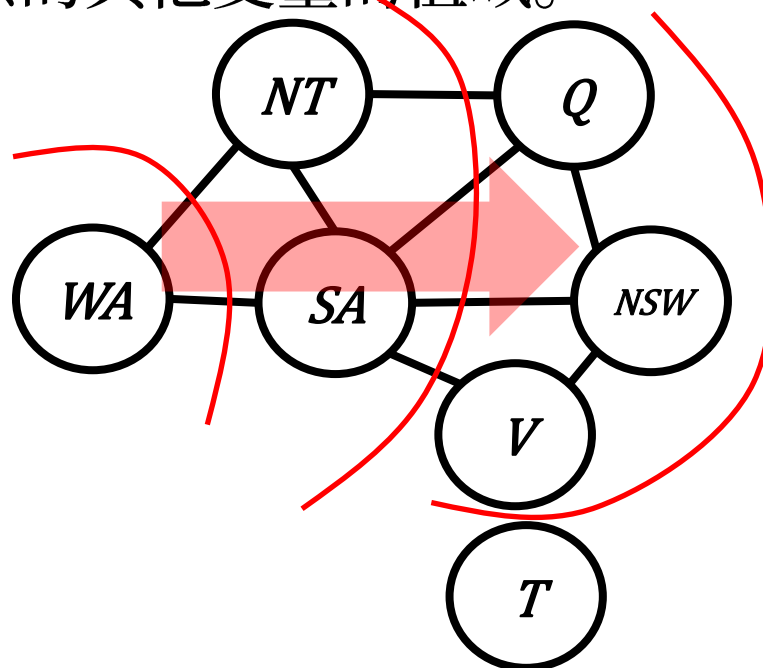
2 约束传播：CSP中的推理



结点相容(1-相容)、弧相容(2-相容)、路径相容(3-相容)、K-相容

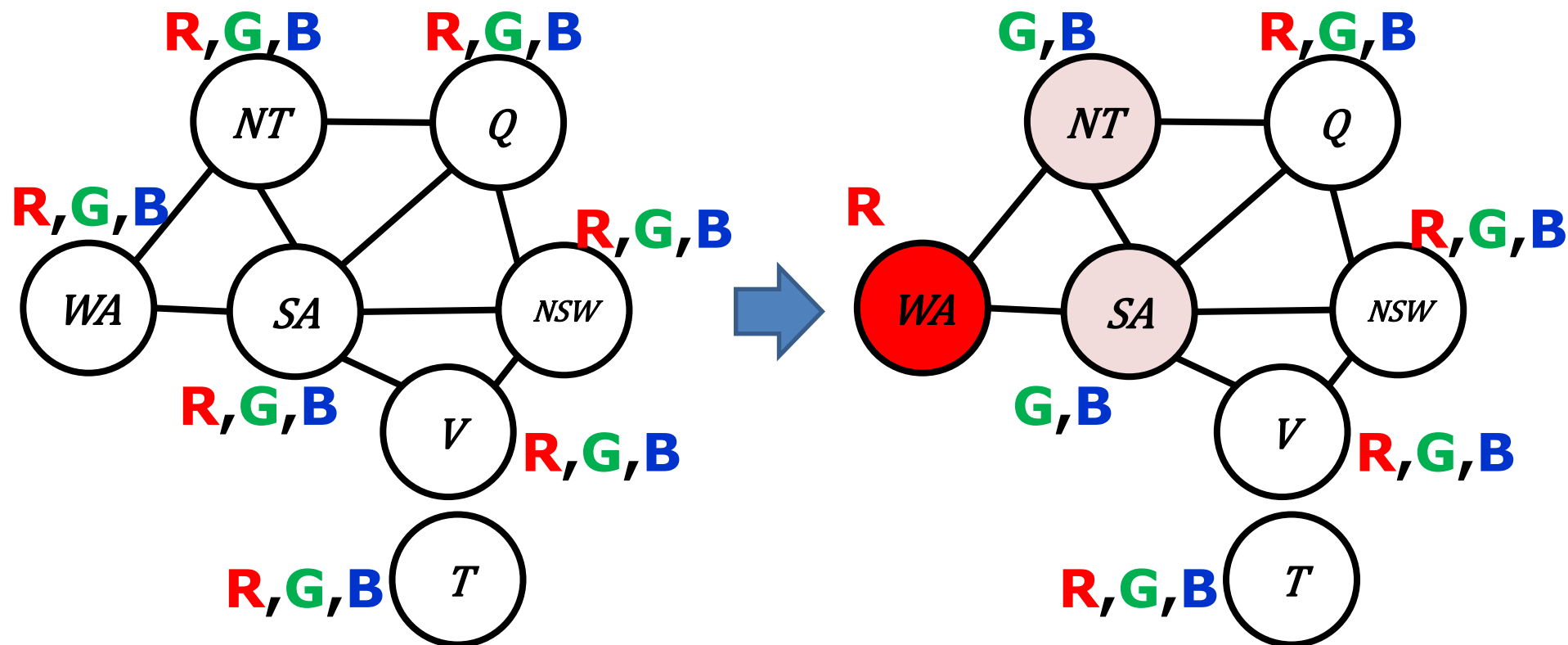
约束传播

- CSP中的推理：约束传播。约束传播使用约束减小一个变量的合法取值范围，从而影响到跟此变量有约束关系的另一变量的取值。
- 给一个变量赋值后，则通过约束传播可缩小与其有约束关系的其他变量的值域。



约束传播

□ 若WA赋值R, 则通过约束传播, NT和SA的值域中删除R



约束传播和搜索

- 在CSP中,算法可以进行**约束传播**,也可以**搜索**
(从几种可能性中选择新的变量赋值)。约束传播与搜索可以**交替进行**,也可以将约束传播作为搜索前的**预处理步骤**。

搜索

约束传播

选择变量
赋值

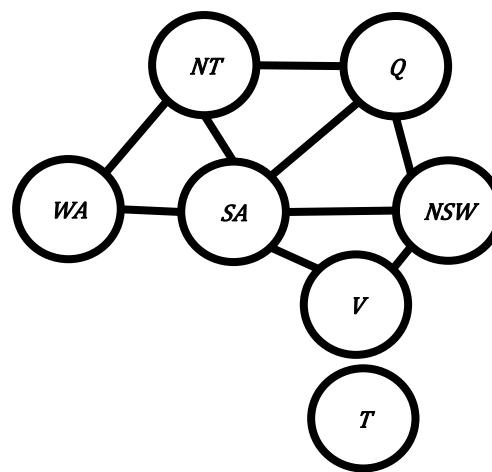
缩小其他
变量值域



约束传播的核心是
通过删除不相容的结点取值增强局部相容性

局部相容性

- 约束传播的核心思想是局部相容性：增强约束图中各部分局部相容性会导致不相容的结点取值被删除。

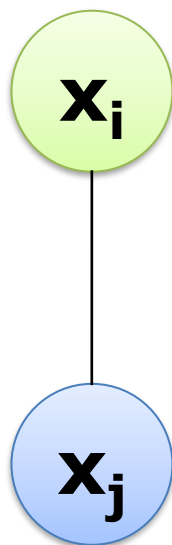


结点相容 Node consistency

- 如果单个变量（对应于CSP网络中的结点）值域中的所有取值满足它的一元约束，就称此变量是结点相容的。
 - 例如，如果地图着色问题中南澳洲人不喜欢绿色，变量 SA 原来值域为 $\{red, green, blue\}$ ，删除 $green$ 此结点即为结点相容的，此时 SA 的值域为 $\{red, blue\}$ 。
- 如果网络中每个变量都是结点相容的，则此网络是结点相容的。

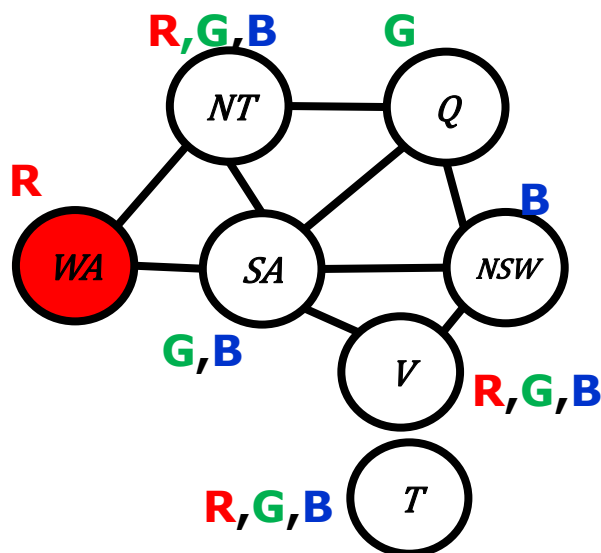
弧相容 arc consistency

- 若对 X_i 的每个赋值， X_j 都存在某个取值满足弧 (X_i, X_j) 的二元约束，则称 X_i 关于 X_j 是弧相容的。



弧相容 arc consistency

- 若对 X_i 的每个赋值, X_j 都存在某个取值满足弧 (X_i, X_j) 的二元约束, 则称 X_i 关于 X_j 是弧相容的。



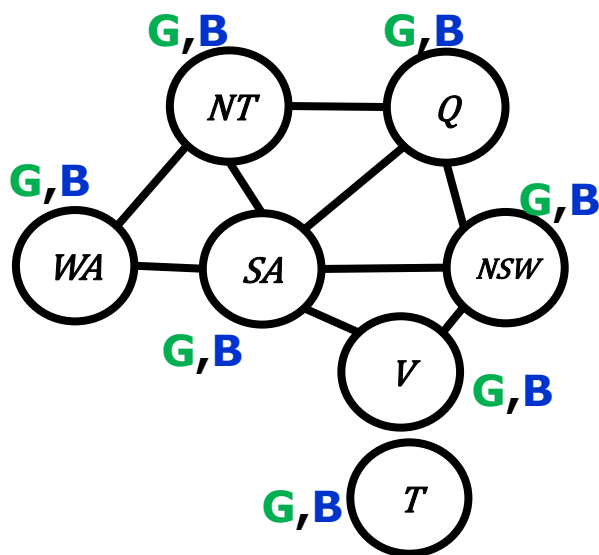
哪些是弧相容的?

- (1) WA关于SA
- (2) WA关于NT
- (3) SA关于Q
- (4) SA关于NSW

弧相容

arc consistency

- 若对 X_i 的每个赋值, X_j 都存在某个取值满足弧 (X_i, X_j) 的二元约束, 则称 X_i 关于 X_j 是弧相容的。
- 如果每个变量关于其它变量都是弧相容的, 则称该网络是弧相容的



网络是弧相容的, 但显然此题无解。(网络是弧相容的, 也不一定有解) 路径相容可以检测出无解。

弧相容算法AC-3

使每个变量关于其他变量都是弧相容的

function AC-3(csp) **returns** false if an inconsistency is found and true otherwise

inputs: csp, a binary CSP with components (X, D, C)

local variables: queue, a queue of arcs, initially all the arcs in csp

while queue is not empty **do**

$(X_i, X_j) \leftarrow \text{REMOVE-FIRST}(\text{queue})$

if REVISE(csp, X_i, X_j) **then**

if size of $D_i = 0$ **then** **return false**

for each X_k **in** $X_i.\text{NEIGHBORS}$ **do**

add (X_k, X_i) to queue

return true

AC-3返回false, 则CSP无解

AC-3返回true, 则CSP不一定有解

CSP有解, 则AC-3一定返回true

Revise(csp, X_i, X_j)之后, X_i 关于 X_j 是弧相容的。返回true, 表示 X_i 的值域被修改过, 否则 X_i 的值域没有被修改过。

function REVISE(csp, X_i, X_j) **returns** true iff we revise the domain of X_i

revised $\leftarrow$ false

for each x **in** D_i **do**

if no value y in D_j allows (x, y) to satisfy the constraint between X_i and X_j **then**

delete x from D_i

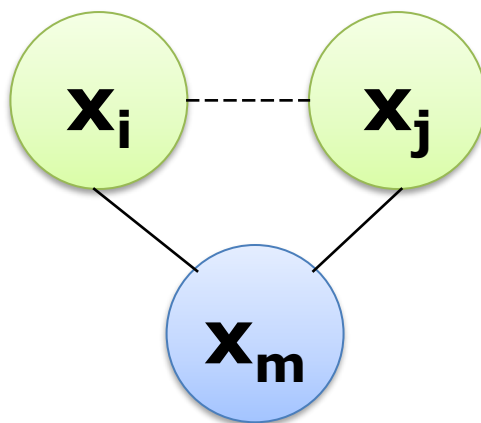
revised $\leftarrow$ true

return revised

路径相容 path consistency

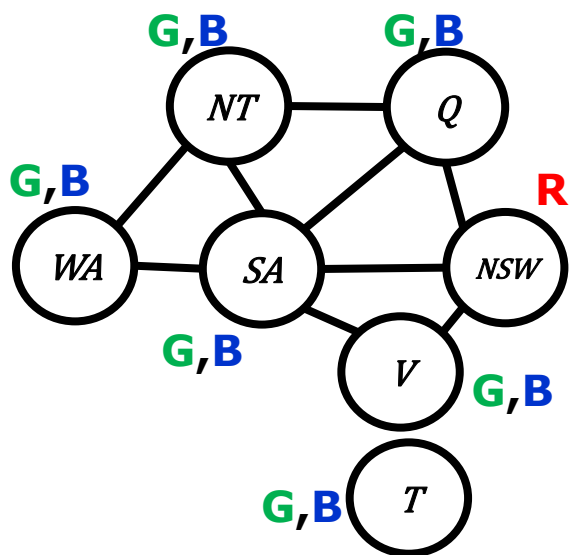
- 对 $\{X_i, X_j\}$ 的每一个相容赋值 $\{X_i = a, X_j = b\}$, X_m 都有合适的取值同时使得 $\{X_i, X_m\}$ 和 $\{X_j, X_m\}$ 是相容的, 则称集合 $\{X_i, X_j\}$ 关于 X_m 是路径相容的。

这很像是一条从 X_i 途经 X_m 到 X_j 的路径



路径相容 path consistency

- 对 $\{X_i, X_j\}$ 的每一个相容赋值 $\{X_i = a, X_j = b\}$, X_m 都有合适的取值同时使得 $\{X_i, X_m\}$ 和 $\{X_j, X_m\}$ 是相容的, 则称集合 $\{X_i, X_j\}$ 关于 X_m 是路径相容的。



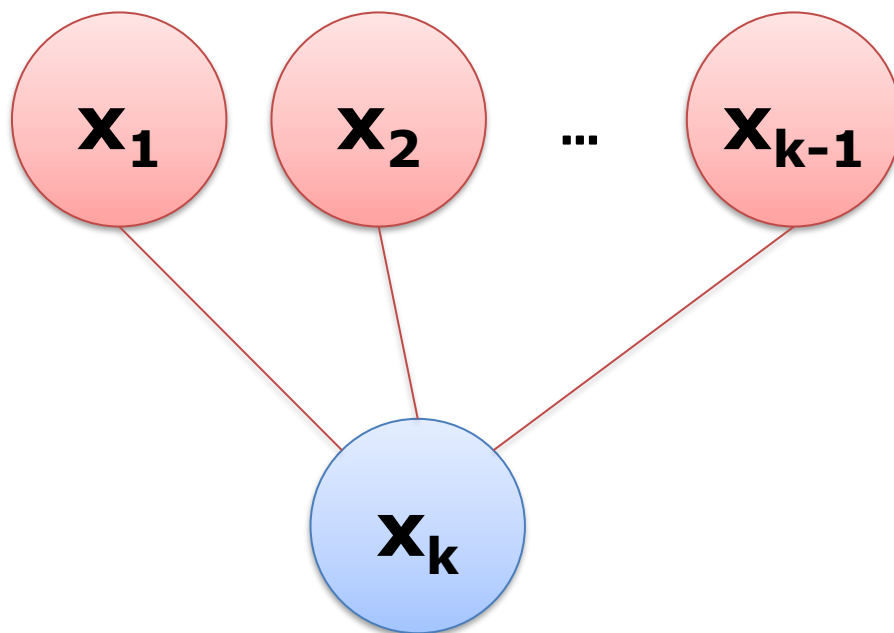
哪些是路径相容的?

- | | |
|------------------------|-----|
| (1) $\{WA, NT\}$ 关于 SA | no |
| (2) $\{SA, Q\}$ 关于 NSW | yes |
| (3) $\{SA, NSW\}$ 关于 V | yes |

k相容

k-consistency

- 给定k个变量 $\{x_1, \dots, x_k\}$, 对于 $\{x_1, \dots, x_{k-1}\}$ 的每一个相容赋值, x_k 都有合适的取值使得 $\{x_1, x_k\}, \dots, \{x_{k-1}, x_k\}$ 是相容的, 则称 $\{x_1, \dots, x_{k-1}\}$ 关于 x_k 是k相容的。



k相容 k-consistency

- 如果对于任何 $k - 1$ 个变量的任何相容赋值，任何第 k 个变量总能被赋予一个和前 $k - 1$ 个变量相容的值，那么称这个CSP是 k 相容的。
- 1-相容是结点相容。
2-相容是为弧相容。
对二元约束网络，3-相容是路径相容。

全局约束的相容性检查

- 全局约束可涉及任意个约束变量（不一定是所有变量）。
- **Alldiff**约束表示所有相关变量必须取不同的值。
*Alldiff*约束的不相容检测的一种简单形式：如果约束涉及 m 个变量，可能的不同取值有 n 个，且 $m > n$ ，那么约束不可能满足。
- 资源约束，有时称为**atmost**约束。例如在调度问题中，用 $P_1, \dots, P_4$ 表示执行四项任务的人数。总人数不超过10人的约束记为 $atmost(10, P_1, P_2, P_3, P_4)$ ：检测值域的最小值之和可以检测相容性。

Thanks
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